

Singly Linked List :

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- 1) b) Data and a pointer to the next node.
- 2) c) NULL
- 3) b) Self-referential ~~the~~ structure.
- 4) a) $\text{head} \rightarrow \text{next} == \text{NULL}$
- 5) b) $\text{newnode} \rightarrow \text{next} = \text{head}$; $\text{head} = \text{newnode}$;
- 6) d) $\text{temp} = \text{head} \rightarrow \text{next}$; ~~free~~ $\text{free}(\text{head})$; $\text{head} = \text{temp} \rightarrow \text{next}$;
- 7) c) Traversing from tail to ~~tail~~ head.
- 8) b) n or $n-1$
- 9) c) Copying the data of $p \rightarrow \text{next}$ into p , then deleting $p \rightarrow \text{next}$
- 10) a) $\text{while}(p \rightarrow \text{next} != \text{NULL})$.
- 11) b) $p \rightarrow \text{next} == \text{NULL}$
- 12) c) Two pointers advancing in different rates.
- 13) b) free
- 14) a) Zero nodes
- 15) b) $p = (\text{struct node} *) \text{malloc}(\text{sizeof}(\text{struct node}))$;
- 16) b) Avoid special case handling.

17) c) The first Node.

18) b) It does not support random (indexed) access

19) b) Less of access to all nodes, causing a memory leak.

20) b) A slow pointer and a fast pointer.

21) a) The head node.

22) b) Node may be scattered anywhere in memory.

23)

```
23) struct Node {  
    int data;  
    struct Node *next;  
};
```

24) 30 50 70

25) ~~A base condition or condition where if (P -~~

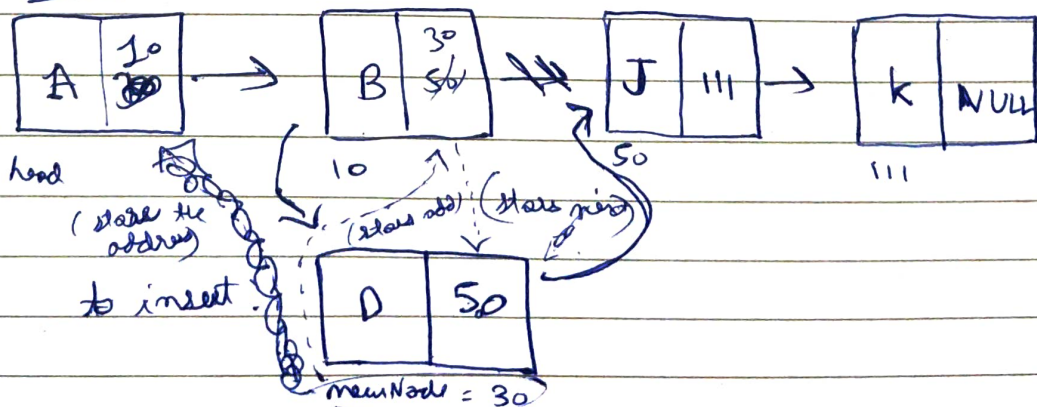
25) ~~A condition is missing that is what if the linked list~~
or

```
25) if ( *head *P == NULL) { // this condition is  
    *p = t; // required  
    return;  
}
```

26)

List 1

27)



- here we've taken a linked list 1, which had the following values
- suppose we want to insert a node b/w any two nodes we need to first create a new Node.
- we've set its ~~address~~ memory address to 30
- Now ~~for~~, we will store the value of the next in temp
 here temp = that node → next :
 hence temp becomes = 50
- Now we'll put the value of ~~that node~~ temp in the new Node's next. (hence it'll point to that node)
- and store the ~~address~~ address of new Node in that 'that node's' next.

28) i) $7 \rightarrow 5 \rightarrow 8 \rightarrow 12 \rightarrow 3 \rightarrow \text{NULL}$

ii) $7 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow \text{NULL}$

iii) $7 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow 9 \rightarrow \text{NULL}$